

Global Financial Uncertainty*

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Abstract

We estimate a novel measure of global financial uncertainty (GFU) with a dynamic factor framework that jointly models global, regional, and country-specific factors. We quantify the impact of GFU shocks on global output with a VAR analysis that achieves set-identification via a combination of narrative, sign, ratio, and correlation restrictions. We find that the contraction in world output during the Great Recession would have been 13% milder in absence of GFU shocks. We also find support for a global finance uncertainty multiplier: the more global financial conditions deteriorate after a GFU shock, the larger the world output contraction is.

Keywords: Global Financial Uncertainty, dynamic hierarchical factor model, structural VAR, world output loss, global finance uncertainty multiplier.

JEL codes: C32, E32.

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1 Introduction

"While all of the recessions since 1985 have had financial origins, the Great Recession was by far the biggest of these. [...] The Great Recession was indeed marked by a high level of uncertainty. [...] The Great Recession was also exceptionally global in nature."

[Serena Ng and Jonathan H. Wright, *Journal of Economic Literature*, 2013]

This paper estimates a novel measure of global financial uncertainty (GFU henceforth) and quantifies the world output loss due to GFU shocks during and in the aftermath of the Great Recession. We proceed in two steps. First, we obtain our GFU index by taking a dynamic hierarchical factor model à la Moench, Ng, and Potter (2013) to the data.¹ Our dataset comprises about 38,000 financial volatility observations spanning the period July 1992 - May 2020, including 42 countries that cover five continents and account for 83% of the world output.² Given the rich geographical coverage of our dataset, which includes heterogeneous countries and regions, we jointly model global, regional, and country-specific uncertainty factors to minimize the risk of attributing spurious dynamics to our global factor. We estimate the factor model by extending the frequentist estimation algorithm proposed by Moench, Ng, and Potter (2013) with the approach put forth by Banbura and Modugno (2014), which is specifically designed to extract factors from datasets like ours characterized by missing observations.

In the second part of the paper, we estimate a VAR model that features our GFU measure, the global financial cycle proposed by Rey (2018) and Miranda-Agrippino and Rey (2020b) (GFC), and the world industrial production index (WIP) estimated

¹Following Bloom (2009), Cesa-Bianchi, Pesaran, and Rebucci (2020), and Baker, Bloom, and Terry (2022), this paper treats realized volatility as a proxy for uncertainty. The use of realized volatility - as opposed to implied (ex-ante) volatility - is due to the lack of official implied-volatility indices for most countries in our dataset. Importantly, these two concepts display a strong correlation at a monthly level - for instance, the correlation between realized and implied financial volatility in the US is 0.90 in our sample, as also noticed by Cesa-Bianchi, Pesaran, and Rebucci (2020). The use of realized volatility as a proxy for uncertainty has recently been criticized by Berger, Dew-Becker, and Giglio (2020), who find modest business cycle effects in response to uncertainty shocks when controlling for realized volatility. However, Alessandri, Gazzani, and Vicondoa (2020) show that the evidence put forth by Berger, Dew-Becker, and Giglio (2020) crucially depends on the assumption of no contemporaneous impact of implied volatility on realized volatility within a month. The identification approach adopted in our paper does not rely on zero restrictions. Our paper does not distinguish between Knightian uncertainty and risk. For a paper empirically dealing with these two concepts, see Rossi, Sekhposyan, and Soupre (2019).

²The list of countries in our database is documented in our Appendix. The data used in this analysis, our estimated factors, and the Matlab codes to replicate our results are available at <https://sites.google.com/site/efremcastelnuovo/home>.

by Baumeister and Hamilton (2019). The first measure helps us separate GFU shocks from first-moment financial shocks, while WIP is a proxy for global output that we focus on to compute the world output loss due to GFU shocks during the Great Recession. We achieve set-identification of GFU shocks via a novel combination of narrative, sign, ratio, and correlation restrictions. Following Ludvigson, Ma, and Ng (2021), we use narrative restrictions and identify key dates - corresponding to large jumps in GFU - in which financial markets were particularly volatile at a world level.³ We impose sign restrictions on our VAR impulse responses to sharpen the identification of GFU shocks and separate them from global financial shocks and "output shocks". As in Furlanetto, Ravazzolo, and Sarferaz (2019) (who work with US data), we disentangle first and second-moment global financial shocks by imposing restrictions on the ratios of the GFU and GFC impulse responses to their respective shocks, i.e., we require a GFU (GFC) shock to increase (decrease) the on-impact response of the GFU/GFC ratio. Finally, we require our GFU shocks to be orthogonal to US monetary policy and oil supply shocks, which have been found to affect - respectively - the global financial cycle (Miranda-Agrippino and Rey (2020b)) and global output (Baumeister and Hamilton (2019)).

Our main results are the following. First, our estimated GFU factor spikes in correspondence with well-known episodes of global financial volatility, chiefly the Great Recession and the COVID-19 pandemic. Second, GFU is highly correlated with regional or country-specific uncertainty of "hegemon" regions such as North America and Europe and countries like the US, the UK, and Germany, but also with uncertainty in small-open economies like Australia and New Zealand. Differently, GFU is much less correlated with uncertainty in Latin America and countries like China, Italy, and Greece that display idiosyncratic dynamics. Third, our GFU index is positively correlated with, but not a perfect substitute for, other measures of US financial uncertainty such as the VIX and the financial uncertainty index recently proposed by Ludvigson, Ma, and Ng (2021). It is also correlated, though to a lesser extent, with estimates of global macroeconomic uncertainty proposed by Redl (2017), Mumtaz and Theodoridis (2017), and Carriero, Clark, and Marcellino (2020), while no correlation is found with measures of global political uncertainty or risk, such as the global economic policy uncertainty index proposed by Baker, Bloom, and Davis (2016) and Davis (2016), the world uncertainty

³Antolín-Díaz and Rubio-Ramírez (2018) propose a narrative-based identification approach that differs from Ludvigson et al.'s (2021) among many dimensions. We discuss these two approaches in Section 5.

index proposed by Ahir, Bloom, and Furceri (2022), and the geopolitical risk measure put forth by Caldara and Iacoviello (2022).

Turning to our VAR analysis, we find GFU shocks to exert a negative impact on both the global financial cycle and global output, with a (median) contribution to global financial cycle's volatility at business cycle frequencies of about 30%, and to that of global output of about 9%. Focusing on the Great Recession and its aftermath, our results show that the loss in WIP would have been 13% lower in magnitude had GFU shocks been absent. Also, our set-identified VAR analysis shows that retained models that imply a large (small) world output loss also imply a mild (severe) deterioration of the global financial cycle. This correlation is consistent with the "finance uncertainty multiplier" hypothesis put forward by Gilchrist, Sim, and Zakrajšek (2014), Arellano, Bai, and Kehoe (2019), and Alfaro, Bloom, and Lin (2021): uncertainty shocks have more disruptive real effects when they also trigger financial stress, with the latter acting as a "magnifier" of the direct negative effects of uncertainty on output.

The paper is organized as follows. Section 2 presents the econometric model. Section 3 describes the data, documents the estimated global financial uncertainty factor, and discusses its properties. Section 4 presents our structural VAR-based analysis. Section 5 discusses the related literature. Section 6 concludes.

2 Data and econometric framework

2.1 Data

Our database features daily data on stock market returns, exchange rate returns, and 10-year Government bond yields.⁴ The focus on these three financial indicators is justified by the following considerations. Stock market volatility has been argued to be an empirically informative proxy for uncertainty (Bloom (2009), Cesa-Bianchi, Pesaran, and Rebucci (2020), Baker, Bloom, and Terry (2022)), and it has been theoretically linked to micro-founded shocks such as shocks to the second-moment of households' discount factor (Basu and Bundick (2017)). Exchange rate uncertainty has been empirically and theoretically shown to be a relevant driver of the optimal currency portfolio formation by investors, with consequences for the business cycle (Benigno, Benigno, and

⁴All series were downloaded from Datastream. Exchange rate data refer to the nominal bilateral exchange rate against the US dollar for all the non-US countries, and to the nominal major currency index for the US. These choices are justified by the role played by the US dollar as the dominant currency on the global markets (Gopinath, Boz, Casas, Diez, Gourinchas, and Plagborg-Møller (2020)).

Nisticò (2012), Gourio, Siemer, and Verdelhan (2013)). Uncertainty surrounding bond yields is naturally linked to the issue of public debt sustainability, which is once again related to portfolio choices and precautionary savings (Fernández-Villaverde, Guerrón-Quintana, Kuester, and Rubio-Ramírez (2015)).⁵ We employ daily data to compute monthly volatilities for the period July 1992 - May 2020. Our unbalanced panel features 122 series for a total of 37,710 observations.⁶

To fix ideas on the importance of modeling our data with a multi-level framework, consider stock market volatility. A simple correlation analysis reveals that: i) in correspondence with the Great Recession and the outbreak of the COVID-19 pandemic, all volatilities series spike upward simultaneously; ii) their average pairwise correlation is 0.44; iii) there is a lot of heterogeneity at a pairwise-correlation level, e.g., stock market volatilities in the US and Canada are very highly correlated (0.91), while the correlation between the volatility of the US and Greek stock market is substantially lower (0.48), and that between the US stock market and China's is basically zero in our sample. We take this *prima facie* evidence (further discussed in our Appendix) as supportive of a multi-level factor approach featuring global, regional, and country-specific components.

2.2 Econometric framework

We model our measures of financial volatility with the dynamic hierarchical factor model (DHFM hereafter) proposed by Moench, Ng, and Potter (2013). Let Z_{rcnt} be the monthly volatility at time t for variable n and country c belonging to region r , where Z_{rcnt} is mean zero and standardized to have unit variance. The four level factor model is given by

$$Z_{rcnt} = \lambda_{C,rc}^n(L) C_{rct} + e_{Z_{rcnt}} \quad (1)$$

$$C_{rct} = \Lambda_{R.rc}(L) R_{rt} + e_{C_{rct}} \quad (2)$$

$$R_{rt} = \Lambda_{GFU.r}(L) GFU_t + e_{R_{rt}} \quad (3)$$

$$\Psi_{GFU}(L) GFU_t = e_{GFU_t} \quad (4)$$

⁵We prefer to include long-term interest rates rather than short-term ones because of the zero (or effective) lower bound in a number of countries during the Great Recession and the COVID-19 pandemic. Data limitation prevents us from dealing with information on financial volumes (credit, money, leverage) in our analysis.

⁶The computation of our financial volatilities is detailed in our Appendix. A balanced dataset conditional on 42 countries, three volatility series per country (financial volatility data on stock prices, exchange rates, and Government bond yields), and 335 observations per series (the number of months in the July 1992-May 2020 sample) would feature 42,210 observations. Hence, the missing observations we have to deal with are 4,500 (42,210-37,710).

where GFU_t is the common factor, $\Lambda_{GFU.r}(L)$ is the distributed lag of loadings on the common factor, $e_{R_{rt}}$ is the region-specific variation, R_{rt} is the $(k_R \times 1)$ vector of region-specific factors, where k_R denotes the number of regions, $\Lambda_{R.rc}(L)$ is the distributed lag of loadings on the region-specific factors, $e_{C_{rct}}$ is the country-specific variation, C_{rct} is the $(K_{Cr} \times 1)$ vector of country-specific factors, where K_{Cr} denotes the number of countries in region r , $\lambda_{C,rc}^n(L)$ is the distributed lag of loadings on the country-specific factors, and $e_{Z_{rcnt}}$ is the series specific variation.⁷ The hierarchical model (1)-(4) is such that only the country-level factors appear in the measurement equations (1). The country-level factors evolve according to a factor model where the common components are the regional factors and, in turn, the regional factors evolve according to a factor model where the common component is the world factor.⁸ All factors are assumed to be stationary, normally distributed autoregressive processes. To handle our unbalanced panel, we estimate the model (1)-(4) by combining the Moench, Ng, and Potter (2013) MCMC algorithm with the EM approach proposed by Banbura and Modugno (2014) (a detailed explanation of our estimation approach can be found in our Appendix). The estimation generates one global, five regional, and 42 country-specific factors. This framework is more parsimonious than alternatives (e.g., Gregory, Head, and Raynauld (1997), Kose et al. 2003, 2008)) because it features coefficient restrictions due to its hierarchical structure, and it allows for the estimation of non-orthogonal regional and country factors (as opposed to "residual" regional and country-specific components orthogonal to the global one). Further discussions on this approach vs. alternatives can be found in Del Negro and Schorfheide (2011) and Moench, Ng, and Potter (2013).

3 Global Financial Uncertainty

3.1 GFU

Features. Figure 1 plots the estimated GFU factor. The index peaks in correspondence with well known historical episodes, which are: the European Monetary System collapse in 1992, the Asian crisis in 1997, the Russian crisis and the LTCM default in 1998, 9/11,

⁷Similarly to financial factors like the NFCI estimated by the Federal Reserve Bank of Chicago and the global financial cycle proposed by Miranda-Agrippino and Rey (2020b), the estimation of GFU involves only financial volatility data and not macroeconomic indicators. This choice is also supported by Cesa-Bianchi, Pesaran, and Rebucci (2020), who document that world growth shocks explain less than 5% of the forecast error variance of global financial volatility.

⁸Our Appendix rewrites the DHFM model presented here to put in evidence that, as pointed out by Moench, Ng, and Potter (2013), the transition equations for the factors at each level feature time-varying intercepts that depend on the factors at the next higher level.

Gulf war II, the US sub-prime mortgage default rate rise, the global financial crisis, the Greek debt crisis and the subsequent Eurozone crisis, the Chinese stock market turmoil, the Brexit referendum in 2016, and the COVID-19 pandemic. Notably, all these peaks are truly global, i.e., not just US-related. The 95% confidence bands of our estimated factor suggest that our GFU measure is quite precisely estimated. This is good news for the VAR exercise we document in the following Section, in that the statistical measurement-error issue potentially affecting our VAR analysis is likely to be of limited importance.

Robustness. Misspecification of the sub-blocks (regions) could in principle have significant effects on the estimated factors (Francis, Owyang, and Savascin (2017)). We tackle this issue by computing two alternative measures of GFU: i) a weighted average of the financial volatility indicators in our dataset; and ii) a common factor estimated by principal components. We find that both measures are highly correlated with our baseline GFU. Another possible concern is the assumption of normality of the measurement and transition shocks in the state-space representation of the DHFM (for a similar choice in the context of financial indicators, see Diebold, Li, and Yue (2008)).⁹ Two observations are in order. First, we notice that the global factor estimated by PCA, which does not assume normality, is very similar to our GFU. Second, we employ two alternative measures of volatility: i) log-volatility, which has been shown to limit the non-Gaussianity issue (above all after accounting for outliers or changes in the factor structure, see Han and Inoue (2015)); and ii) realized bipower variation, which is robust to the presence of jumps (Barndorff-Nielsen and Shephard (2004)). Our Appendix offers a detailed description of how these alternative factors have been constructed and shows that they are all highly correlated with our baseline GFU.

Comparison with financial indicators. Figure 2 (upper panel) plots our GFU proxy, the VIX (one of the most popular measures of financial uncertainty in the literature), the US financial uncertainty index by Ludvigson, Ma, and Ng (2021), the global financial cycle produced by Miranda-Agrippino and Rey (2020b), and the National Financial Conditions index (NFCI) produced by the Federal Reserve Bank of

⁹One could interpret our factor model as an approximating model of the true data-generating process and undertake a quasi-maximum likelihood estimation of the factors under misspecification (see Sentana and Shah (1994) for a case concerning static factors, and Doz, Giannone, and Reichlin (2012) for an investigation of the properties of the quasi-maximum likelihood estimator for dynamic factor models in large panels with cross-correlated idiosyncratic components). These frameworks typically assume the existence of only two layers, i.e., common factors and idiosyncratic components. Unfortunately, maximum-likelihood would hardly be feasible to estimate a four-layer model like ours.

Chicago.¹⁰ The scatterplots proposed in Figure 2 (bottom panels) show that the GFU index correlates positively with all three measures, pretty strongly with the VIX (0.86) and Ludvigson et al.'s (2021) US financial uncertainty index (0.68). Given the dominant role played by the US economy in the world financial markets, this is not a surprise. The correlation between GFU and the NFCI is also high (0.74). This can be thought of as a manifestation of the financial conditions-volatility toxic tandem that often materializes in periods of financial stress (Caldara, Fuentes-Albero, Gilchrist, and Zakrajšek (2016), Alessandri and Mumtaz (2019), Alfaro, Bloom, and Lin (2021), Alessandri and Bottero (2020), Caggiano, Castelnovo, Delrio, and Kima (2021), Brianti (2021), and De Santis and Van der Veken (2021)).¹¹ GFU also correlates with the GFC factor, but the degree of correlation is much lower (0.33). This points to the possibility of separately isolating GFU and GFC shocks in our VAR analysis, an issue we deal with in the next Section.

GFU vs. VIX. While co-moving, our GFU factor and the VIX behave differently from a quantitative standpoint in correspondence with the extreme event we focus on in this paper, i.e., the Great Recession. World output dramatically dropped about 14% in one year from 2008M1 to 2009M1. GFU skyrocketed in that period, recording in 2008M10 (the month in which it reaches its peak) a value 5.5 larger than that in 2008M1. The VIX also dramatically hiked up during the Great Recession. However, with respect to 2008M1, its value was "just" 2.5 times larger in 2008M11, the month when it peaked during the global financial (and real) crisis. This makes sense: our GFU factor is constructed to capture financial uncertainty at a global level, while the VIX - often interpreted as a proxy for global financial uncertainty - is chiefly connected with the US financial markets. Experiments conducted with VARs modeling world output, the global financial cycle, and either the GFU or the VIX returns a contribution of GFU shocks to the global real and financial volatility larger than that implied by shocks to the VIX (for a similar finding, see Cesa-Bianchi, Pesaran, and Rebucci (2020)).¹² Hence,

¹⁰Ludvigson et al.'s (2021) US financial uncertainty factor is the time-varying volatility of the one-step ahead forecast errors related to 148 monthly financial series and computed over the period 1960-2020. Differently, Miranda-Agrippino and Rey (2020b) compute a factor with a principal component approach on 1,002 series of asset prices traded on all the major global markets, a collection of corporate bond indices, and commodities price series over the sample 1990 to 2019. They find that such global financial factor (GFC) explains about 20% of the variance in the data. The indices are normalized to have the GFU's mean and variance. The global financial cycle (GFC) index displayed in Figure 2 has its sign flipped to ease comparisons – so that an increase in GFC in the figure should be interpreted as a deterioration of global financial conditions.

¹¹Part of this correlation is likely due to the presence of volatility indicators (e.g., the VIX, or the crude oil volatility index) among those employed to estimate the National Financial Conditions index. The full list of indicators can be found at <https://www.chicagofed.org/research/data/nfci/background>.

¹²A recursive VAR (which we employed to have a "first-pass" of the data) that orders a measure of

while the VIX can be seen as a proxy for the GFU, the two are hardly perfect substitutes.

Comparison with existing global uncertainty factors. Figure 3 plots our GFU index against six different measures of global uncertainty/risk: Davis' (2016) global economic policy uncertainty, the world uncertainty index (WUI) by Ahir, Bloom, and Furceri (2022), and the geopolitical risk (GPR) index by Caldara and Iacoviello (2022), all constructed with a text search-approach following the lead by Baker, Bloom, and Davis (2016) (newspapers' as for GEPU and GPR, Economist Intelligence Unit country reports' as for WUI); as well as Redl's (2017), Muntaz and Theodoridis' (2017) and Carriero, Clark, and Marcellino's (2020) measures of global macroeconomic uncertainty. The degree of correlation between GFU and the three measures of macroeconomic uncertainty is positive, ranging from 0.59 to 0.71. Notably, both our and their measures peak in correspondence with the Great Recession. However, the macro measures display less distinct peaks with respect to our financial volatility series, a difference which is justified by the slower-moving behavior of macroeconomic indicators with respect to financial ones. Much lower correlations emerge when we compare our GFU measure with the text-based ones (the correlation of GFU with GEPU reads 0.14, while with WUI and GPR is basically zero). This is in part due to the fact that GFU features its global maximum during the Great Recession, while the text-based measures do not; that the newspaper-based uncertainty has gradually taken off since 2016, while GFU - while picking up the Brexit referendum - displays a dramatic jump in 2020; and by the fact that GPR peaks in correspondence with the US invasion of Iraq in March 2003. Such measures point to a persistently high uncertainty at the end of the sample, while our factor (as typical of financial volatility proxies) is characterized by a lower degree of persistence. All in all, we interpret these correlations, which range from zero to 0.71, as supportive of the genuinely new information on global uncertainty carried by GFU.

Correlation between GFU and regional/country-specific factors. GFU is highly correlated (about 0.9) with regional factors such as those of North American, Europe, and Oceania, as well as those of countries such as the US, the UK, Germany, Australia, and New Zealand. The correlation between GFU and the North American factor is arguably driven by the role played by the US on the international financial markets, and by its strong influence on many countries around the world, and obviously within North America (for evidence on spillovers of US uncertainty shocks onto the

world output first, a measure of global financial cycle second, and either the GFU or the VIX third returns a contribution to the forecast error volatility of world output (2-year ahead horizon, sample: 1992M7-2019M4) of about 18% if the GFU is used as a proxy for global financial variance vs. just 2.35% when the VIX is considered.

Canadian economy, see Caggiano, Castelnovo, and Figueres (2020)). UK and Germany are influential players on the European financial market, with spillovers on the global scenario. Differently, Oceania is a textbook example of open economy region buffeted by external shocks. The estimated financial uncertainty factors associated with Asia and Latin America, as well as China, Japan, and Singapore, are characterized by a much lower correlation. This is likely to be due to the role played by idiosyncratic events such as the tax reforms that generated uncertainty in China in the early 1990s, the crisis triggered by the collapse of the Thai Baht in 1997 for the case of Asia, and the Peso and debt crisis in Argentina and Mexico. Italy and Greece feature financial volatility factors less synchronized with respect to the rest of Europe and the world as a whole, a fact possibly related to the exchange rate crisis that affected Italy in the early 1990s and the debt crisis in Greece in 2012. Our Appendix adds further details and discussion on the estimated regional and country-specific factors.

4 VAR Analysis

Reduced-form VAR. We consider the vector $\mathbf{X}_t = [GFU_t, GFC_t, 100 \log(WIP_t)]'$, where GFU is our global financial uncertainty index, GFC is the global financial cycle estimated by Miranda-Agrippino and Rey (2020b), and WIP stands for the level of world industrial production developed by Baumeister and Hamilton (2019).¹³ The first two variables are modeled in levels, while WIP - as indicated above - is modeled in logs and multiplied by one hundred.¹⁴ The monthly world industrial production index covers OECD countries and six major emerging markets (Brazil, China, India, Indonesia, the Russian Federation and South Africa). It measures the physical volume of output produced in the industry sector. Originally constructed by the OECD and recently updated by Baumeister and Hamilton (2019), it has been used to conduct empirical investigations on the drivers of global indicators (see Baumeister and Hamilton (2019) for the role of oil shocks as drivers of world output, and Miranda-Agrippino and Rey

¹³Small-scale VARs to investigate the drivers of the world business cycle have been employed by, among others, Kilian (2009), Kilian and Murphy (2012, 2014), Antolín-Díaz and Rubio-Ramírez (2018), and Baumeister and Hamilton (2019).

¹⁴We model world output in log-levels (as opposed to growth rates) to minimize problems related to overdifferentencing. Canova (2007) suggests to model variables in log-levels and check if residuals are stationary, which is indeed the case for our reduced-form VAR. Reassuringly, in this case consistent estimates of VAR coefficients obtain with classical methods even when unit roots are present. Further discussions on data modeling and transformation in VAR investigations can be found in Canova (2007, pp. 124-125).

(2020b) for that of US monetary policy shocks). As documented by Baumeister, Korobilis, and Lee (2022), forecasting exercises involving a variety of global indices point to WIP (along with the real commodity prices factor) as the most informative indicator to predict global variables such as real oil prices and global petroleum consumption.¹⁵

Our VAR analysis covers the sample 1992M7-2019M4. Its span is dictated by the availability of the GFC and GFU series.¹⁶ The VAR features three lags as suggested by the Schwarz information criterion.¹⁷

Identification. Set-identification of the GFU shocks is achieved by searching for rotations of the Cholesky factor of the reduced-form VAR covariance matrix that imply structural models consistent with our identification constraints, whose presentation comes next.¹⁸

Event constraints. Event constraints are constraints imposed directly on the shocks implied by each rotation (Antolín-Díaz and Rubio-Ramírez (2018), Ludvigson, Ma, and Ng (2021)). We impose constraints on dates that are associated with large jumps in financial uncertainty that have a clear interpretation from an historical standpoint. To identify such dates, we proceed as follows. First, we compute the median values over time of the GFU shocks corresponding to $K = 500,000$ admissible models, i.e., models equally consistent with the data. Second, we identify dates associated with a value larger than two standard deviations of the time series of the median values itself. This algorithm returns six dates associated with historical events characterized by high financial volatility: the Russian crisis and LTCM default in August 1998, 9/11, the Worldcom and Enron scandals in June 2002, the acceleration of the Great Recession due to Lehman Brothers' bankruptcy in September 2008, the Greek crisis in April 2010, and the Eurozone crisis in August 2011. Our event constraints require that rotations

¹⁵Baumeister, Korobilis, and Lee (2022) also propose a new index of global economic conditions - termed "GECON" - by combining various indicators of the global business cycle. The correlation between the growth rate of WIP and GECON in our sample is 0.63. Baumeister and Hamilton (2019) and our Appendix offer further details on the construction of WIP.

¹⁶The exclusion of COVID-19 data minimizes the risk of distortions of the VAR coefficients and impulse responses due to the presence of a few outliers (Lenza and Primiceri (2022)). One can notice that, while the VAR analysis excludes the COVID-19 period, the DHFM model we use to estimate our global financial factor includes them. Our Appendix contrasts the estimate of the GFU factor obtained by including vs. omitting COVID-19-related observations, and shows that the estimated GFU factor in the pre-COVID-19 sample is basically the same. For a recent paper computing financial (and macroeconomic) uncertainty for the US with a factor approach that models the extreme volatility of the COVID-19 observations, see Carriero, Clark, Marcellino, and Mertens (2020).

¹⁷Our Appendix shows that our impulse responses are robust to the employment of 5 lags, which is the lag structure suggested by the Akaike information criterion. For an investigation on the selection of VAR lags, see Ivanov and Kilian (2005).

¹⁸Details on the reduced-form - structural VAR mapping are provided in our Appendix.

(structural models) must display "large" realizations of GFU shocks on these dates, i.e., they have to be larger than the median of the distribution of the shocks across all models on these very same dates, with the exception of the Great Recession case, for which our requirement is that GFU shocks realizations have to be larger than the 75th percentile of such distribution.¹⁹ As a matter of fact, this way of searching for "large shocks" is data-driven, i.e., no a-priori restriction based on economic theory is involved in this step.

Sign and correlation constraints. To separate GFU shocks and global "output shocks", we impose two different sets of restrictions. First, we require "output shocks" to be consistent with the idea of a global downturn during the December 2007 - June 2009 Great Recession, which is the dating of the US recession by the NBER. Following Ludvigson, Ma, and Ng (2021), we impose that the sum of the realizations of the output shocks in that period must be negative. Second, we select other potential shocks that, according to the extant literature, might act as drivers of the world business cycle and require that the GFU shocks must be uncorrelated with them. Miranda-Agrippino and Rey (2020b) find US monetary policy shocks to be followed by significant movements in their global financial cycle, while Baumeister and Hamilton (2019) identify oil supply shocks as the most important driver of their measure of world industrial production. To minimize the risk of identifying convolution of shocks rather than the primitive shocks of interest (see Canova and Ferroni (2021)), we require our models to return a correlation between GFU shocks and Miranda-Agrippino and Rey's (2020) US monetary policy shocks on the one hand, and GFU shocks and Baumeister and Hamilton's (2019) shocks on the other, lower than 0.05.²⁰

Ratio and further constraints. A notoriously challenging task is that of empirically separating first and second-moment financial shocks (for a discussion, see Stock and Watson (2012)). Events such as Lehman Brothers' bankruptcy or the Euro area-related crises are likely to be associated with uncertainty shocks but also with shocks to financial conditions. To separate these two different shocks, we follow Furlanetto, Ravazzolo, and

¹⁹A chart with the median realizations of the shocks and the identified dates is available in our Appendix. Our results are robust to requiring that GFU shocks realizations be larger than the 50th percentile of the reference distribution in 2008M9.

²⁰We do not require this correlation to be exactly zero for two reasons. First, our analysis is necessarily a (limited) sample analysis (as opposed to a population analysis). Second, US monetary policy and oil supply shocks are estimated objects surrounded by statistical uncertainty. An alternative to this way of implementing the correlation restrictions is to discard all models with estimated correlations that are not significantly different from zero at a given significance level. Our results are robust to working with this alternative restrictions (significance level: 5%).

Sarferaz (2019) (whose application is on US data) and impose the requirement that a GFU (GFC) shock causes an on-impact positive (negative) response of the GFU/GFC ratio.²¹ We assume both first-moment GFC shocks and second-moment GFU shocks to deteriorate the global financial cycle and increase global uncertainty. However, we require the former shock to generate a larger on-impact response of the global financial cycle than of global uncertainty. The idea is that a first-moment financial shock has a quantitatively larger on-impact effect on the global financial cycle than on global uncertainty.²² Notice that the sign of the GFU/GFC ratio is endogenously determined over horizons - it is constrained only on impact. We also assume a global uncertainty shock to generate an on-impact increase in the GFU/GFC ratio because of the larger quantitative on-impact effect on GFU than on GFC. Finally, we require a positive global financial (output) shock to be expansionary (generate a positive response of the global financial cycle) on impact. Notably, imposing these restrictions does not imply any explicit or implicit assumption of recursiveness of the economy.

The set of constraints imposed in our analysis is collected in Table 1. The joint imposition of the above described constraints on the set of unconstrained models $\mathcal{B}$ delivers a set of 406 (0.08% of K) admissible solutions $\bar{\mathcal{B}}$, which we employ to study the macroeconomic effects of GFU shocks. Table 1 also reports the impact of each different type of restriction on the unconstrained set $\mathcal{B}$. We notice that the most selective restrictions are the correlation constraints, which discard about 93% of the models of the unconstrained set $\mathcal{B}$, followed by the event-based restrictions, which end up retaining 11.21% of the 500,000 models of our unconstrained set. All our identification restrictions work in favor of shrinking the set $\mathcal{B}$.

Alternative approaches to identify uncertainty shocks recently proposed by the literature are those based on instruments (e.g., Carriero, Mumtaz, Theodoridis, and Theophilopoulou (2015), Piffer and Podstawski (2018), Alessandri, Gazzani, and Vicondoa (2020)) and those based on heteroskedasticity (Angelini and Fanelli (2019), Angelini, Bacchiocchi, Caggiano, and Fanelli (2019)). With respect to these approaches, which imply point-identification, our set-identification approach allows us to deal with different interpretative models of the data (all equally likely from a statistical stand-

²¹To impose meaningful sign restrictions on the responses of the GFU/GFC ratio, we adjust the GFC series so that its first two moments are equivalent to those of the GFU series.

²²While being intuitive, this restriction may be questionable because changes in volatility, possibly related to macro fundamentals, could be the main driver of larger movements in asset prices (see, e.g., Kim and Kim (2019)). Our Appendix documents that such a restriction is not crucial for our results, which are robust to dropping it.

point). This will enable us to put in evidence the correlation between financial stress and output loss at a global level conditional on our GFU shocks, i.e., the global finance uncertainty multiplier.

4.1 VAR findings

IRFs and FEVD. Figure 4 plots the impulse response functions of global financial uncertainty, the global financial cycle, and world industrial production to a one standard deviation GFU shock.²³ Following Baumeister and Hamilton (2020), we plot the impulse responses implied by all our retained draws to fully account for model uncertainty. A few facts emerge. First, the response of GFU to its own shock peaks around two months after the shock, then goes gradually back to zero within one year. Second, the response of GFC is hump shaped (as the GFU and WIP ones), but takes longer to go back to zero (about two years).²⁴ Third, the response of WIP peaks around ten-to-twelve months after the shock, and goes back to zero after about 2 and 1/2 years. Notably, the peak response of GFC and WIP is clearly negative for all retained models. Model uncertainty implies different point estimates of the peak response of world industrial production to a GFU shock. Panel [2,2] in Figure 4 plots the histogram of such peak response, which is constructed by collecting the maximum negative value of the impulse response of each model. The median realization of this distribution is -0.27, i.e., world industrial production falls by 0.27% in response to a one standard deviation jump in global financial uncertainty, while the set of retained models points to a drop in world output ranging from 0.61% down to 0.09%.²⁵ This evidence is close to that proposed by Cesa-Bianchi, Pesaran, and Rebucci (2020), who also find a persistent and strong effect of global financial volatility shocks on output growth.

²³Our VAR analysis is conducted under the assumption of linearity of the relationships among real and financial cycles. While representing a standard modeling framework, a linear VAR could importantly underestimate the world output effects of a global financial uncertainty shock in presence of global financial stress (for evidence based on US data, see Alessandri and Mumtaz (2019)). An investigation of the nonlinear effects of global financial uncertainty shocks conditional on the state of the global financial markets is an interesting question for future research.

²⁴This hump-shaped response is a sign in favor of the successful separation between GFU shocks and US monetary policy shocks, which - as shown by Miranda-Agrippino and Rey (2020b) - exert their maximum effect on GFC on impact.

²⁵We report the impulse responses corresponding to the full set of identified draws for two reasons. First, as anticipated in the Introduction, we will exploit all retained models to compute the output loss due to GFU shocks during the Great Moderation and the correlation existing between such loss and the impact of GFU shocks on the global financial cycle (what we call "global finance uncertainty multiplier"). Second, as pointed out by Baumeister and Hamilton (2020), reporting the full set of retained draws is a way to investigate how model uncertainty affects our set-identified responses.

Table 2 collects the figures regarding the forecast error variance decomposition at a 2 and 4-year forecast horizon. The contribution of GFU shocks to the world business cycle is around 9%, just slightly lower than that by global financial cycle shocks and comparable to the one found by Cesa-Bianchi, Pesaran, and Rebucci (2020) on the impact of global financial volatility shocks on output growth. The global financial cycle is found to be driven by both GFU shocks (about 30%) and business cycle shocks (about 45%). GFU is mostly affected by its own shocks, but responds also to world output shocks (about 22-27%, depending on the horizon one considers) and global financial shocks (10%).

Non-recursive vs. recursive identification strategy. Figure 4 also compares the impulse responses implied by our non-recursive identification strategy with those generated via a recursive VAR. As for the latter, we order GFU last to control for business cycle and first-moment financial shocks. The Figure reveals the following: i) by construction, the GFU shock in this recursive economy exerts a zero on-impact effect on the global financial cycle. Differently, our non-recursive strategy points to a negative on-impact effect according to all our retained models; ii) as a consequence of i), the point-identified impulse response produced by the recursive VAR features a smaller peak (negative) response of GFC than what suggested by most our retained models; iii) the 95% confidence bands associated with our point-identified responses (generated by the recursive VAR) do not contain all our set-identified responses for a non-negligible number of horizons. This evidence lines up with the one by Ludvigson, Ma, and Ng (2021), who label recursive VARs as "inappropriate" when it comes to studying the business cycle effects of uncertainty shocks (in their case, with US data).²⁶ Our Appendix documents the "short-run" forecast-error variance decomposition over a 3-month horizon, which reveals that relative to recursive identification our non-recursive identification strategy assigns a larger contribution of GFU shocks to the forecast error variance of both the global financial cycle (about 28% vs. 19%) and to that of world output (3.45% vs. 1.66%). Hence, while displaying some similarities in shape and magnitude, the impulse responses produced with our non-recursive identification strategy also suggest a different contribution - at least at a short-horizon - of our identified shocks to the volatility of the global indicators we model with respect to a recursive VAR.

An alternative global financial cycle index. Our GFU factor is estimated

²⁶Our Appendix shows that the on-impact response of GFU to a GFC shock is also non-zero. Hence, while orderings of the variables alternative to the one proposed in the text are possible, no recursive VAR would be able to properly capture the GFU-GFC contemporaneous interconnection.

with data on stock market, exchange rate, and 10-year government bond yields for 42 countries covering five continents (details provided in Section 2). The GFC, first estimated by Miranda-Agrippino and Rey (2015) and Miranda-Agrippino and Rey (2020b), is instead based on an unbalanced panel of 1,002 time series that cover equity prices from five geographical areas (Europe, North America, Latin America, Asia Pacific, and Australia), commodity prices excluding precious metals, and corporate bond indices. (Further details on the construction of the GFC can be found in Nenova and Rey (2020), Miranda-Agrippino and Rey (2020a), and Miranda-Agrippino and Rey (2021).) While being partly overlapping, the two datasets used to construct our measure of global uncertainty and the global financial cycle factor differ along a few dimensions (mainly, geographical representation and covered assets).

We check if our results are robust to employing a measure of the global financial cycle more consistent with our GFU by proceeding in two steps. First, we use our data on financial returns (as opposed to volatility of returns) and estimate a novel (first-moment) financial conditions factor with our dynamic factor model by imposing the very same structure we worked with when we estimated the GFU factor. The correlation between our novel global financial conditions factor - plotted in our Appendix for brevity - and the GFC by Miranda-Agrippino and Rey (2020b) is 0.51. Second, we replace Miranda-Agrippino and Rey's global financial cycle in our VAR with our novel global financial conditions factor and re-estimate the effects of GFU shocks (as well as the other two identified shocks) on global uncertainty, global financial conditions, and world output. Our Appendix reports the estimated impulse responses, which confirm in full that: i) a GFU shock leads to a global financial and real bust; ii) a positive GFC shock induces a temporary but persistent real boom, while not much of a response of GFU is recorded; iii) a world output shock brings along an improvement of global financial conditions but does not have a clear effect on global financial uncertainty.

Global financial vs. macro uncertainty shocks. Our ultimate goal is that of quantifying the world output effects of global financial uncertainty shocks. However, GFU could in principle be driven by macroeconomic uncertainty shocks. We conduct an exercise to investigate if we are confounding the two shocks. We estimate a bivariate reduced-form VAR(3) that models the two measures of financial and macroeconomic uncertainty recently proposed by Jurado, Ludvigson, and Ng (2015) and Ludvigson, Ma, and Ng (2021), and interpret the VAR residuals as proxies for, respectively, US financial uncertainty shocks and US macroeconomic uncertainty shocks. Given the role of the US as "hegemon" on the international financial markets, a relatively safe assumption

is that, if we are not confounding financial and macro uncertainty shocks, our GFU shocks should be correlated (uncorrelated) with proxies for financial (macroeconomic) uncertainty shocks originating in the US. We then check the correlations between these two VAR proxies for US uncertainty shocks and each one of the GFU shocks implied by our 406 retained models. We find that: i) the median correlation between our GFU shocks and the proxy for US financial uncertainty shocks is 0.31, as opposed to the 0.20 we get when considering the proxy for the US macroeconomic uncertainty shocks;²⁷ all 406 models (i.e., all 406 different representations of GFU shocks) display a significant correlation at a 5% level with the proxy for US financial uncertainty shocks, while just 43 out of 406 models (10.59%) feature a significant (5% level of significance) correlation with the proxy for US macroeconomic uncertainty. While not being definitive, this evidence - further documented in our Appendix - is consistent with the interpretation of GFU shocks as financial shocks.

Global vs. regional uncertainty shocks. One way to appreciate the advantages of a multi-layer factor model is that we can test whether country- or region-specific shocks have global effects, or more generally whether global shocks have similar effects to region/country-specific shocks. If we are isolating proper region-specific uncertainty indices, one should expect that uncertainty shocks specific to regions that display financial volatility developments that are different from the global component would have no impact on global output. To verify this conjecture, we estimate a VAR in which we replace our GFU factor with the financial uncertainty factor we estimated for Latin America (which we take as a reference region characterized by shocks different from the global shocks analyzed here). Reassuringly, our results point to a negligible role played by Latin American financial uncertainty shocks for the dynamics of the global financial cycle and world output. Our Appendix offers further details on this "placebo" test.²⁸

²⁷Cesa-Bianchi, Pesaran, and Rebucci (2020) document a strikingly similar correlation (0.32) between their measure of global financial volatility and the change in the US financial volatility proxy proposed by Ludvigson, Ma, and Ng (2021).

²⁸Baumeister and Hamilton (2020) recommend to explore robustness along other two dimensions: i) choice of the seed that initializes the stochastic simulations one conducts when searching for models meeting the identifying constraints; ii) the use of the Haar distribution on rotation matrices $\mathbf{Q}$, which can imply non-uniform distributions over objects such as impulse responses, or the finance-uncertainty multiplier. Our Appendix documents the robustness of our results to the use of different seeds and the impact of our identifying restrictions on the impulse responses and the finance uncertainty multiplier documented in this paper.

4.2 GFU shocks and the Great Recession

How large was the contribution of GFU shocks to the Great Recession? To address this question we use our VAR to simulate - conditional on each retained model - the evolution of WIP and GFC in a counterfactual scenario in which GFU shocks are set to zero during the 2008M9-2012M12 global downturn (Kose and Ohnsorge (2019)). We then quantify the "World Industrial Production Great Recession loss" by computing the distance between the counterfactual world industrial production that (according to our VAR) we would have observed in absence of GFU shocks and the actual one. Formally, we compute:

$$\mathcal{L}_{WIP|e_{GFU.GR}} = \frac{1200}{52} \sum_{t=2008M9}^{2012M12} \log \left(\frac{WIP_t|e_{GFU,t=0}}{WIP_t} \right). \quad (5)$$

The factor $\frac{1200}{52}$ in (5) rescales the cumulative loss during and after the Great Recession by the number of months (52) in the 2008M9-2012M12 period, then multiplies it by 1200 to make the loss interpretable as annualized percent deviation of the world industrial production index in absence of GFU shocks with respect to the actual WIP.

Figure 5 (left panels) plots the evolution of the simulated path that WIP would have followed if no GFU shocks had materialized (and contrasts it with the actual WIP series) according to two of our retained models, i.e., the one that implies the minimum output loss during the period under analysis and the one that implies the maximum loss. The two different scenarios implied by these two different structural models offer strikingly contrasting indications. In one case (that of the minimum loss), the contribution of the GFU shocks appears to be small, if not negligible. The opposite conclusion can be drawn when looking at the maximum loss case, where GFU shocks emerge as relevant drivers of the severe drop and slow recovery of the world economy experienced in the period under scrutiny. The distribution of the $\mathcal{L}_{WIP|e_{GFU.GR}}$ statistic (5), constructed by plotting the histogram of the values of such statistic conditional on all retained models $\bar{B}$ and shown in the upper-right panel of Figure 5, point to a loss ranging from 3% to 27%, with a median value of 13%. This evidence, which suggests that GFU shocks are likely to have played a quantitatively relevant role during the Great Recession, is in line with recent estimates put forth by Cesa-Bianchi, Pesaran, and Rebucci (2020), who find that the combination of negative output growth shocks and recessionary global financial volatility shocks is able to explain an output decline of about 3% (out of a total of 3.5%) in 2008Q4 and 2009Q1.

4.3 Global finance uncertainty multiplier

What does our model suggest as for the financial costs due to GFU shocks during the Great Recession? Figure 5 (middle panels) plots the counterfactual paths that the GFC would have followed in absence of GFU shocks. It is important to stress that the counterfactual simulations of both WIP (Figure 5, left panels) and GFC (Figure 5, middle panels) correspond to the models that imply on the one hand the lowest world industrial production loss according to our set of retained rotations $\bar{\mathcal{B}}$, and on the other hand the highest. A correlation seems to arise. When the output cost due to GFU shocks is small, the impact of GFU shocks on the global financial cycle is also small. Instead, when the loss in world industrial production is large, the financial cycle is also found to be substantially deteriorated.

Is the correlation between world industrial production and global financial cycle losses robust across all our retained models $\bar{\mathcal{B}}$? To address this question, we first compute per each retained model the global financial cycle loss occurred during the Great Recession and due to GFU shocks. We quantify such loss as follows:

$$\mathcal{L}_{GFC|e_{GFU.GR}} = \frac{12}{52} \sum_{t=2008M9}^{2012M12} (GFC_t|e_{GFU,t=0} - GFC_t) \quad (6)$$

We then scatter-plot the values of the losses (5) and (6) across the set of retained models $\bar{\mathcal{B}}$ to see if there is any correlation between these two objects. The bottom-right panel of Figure 5 shows that this correlation is clear in the data. This correlation is consistent with the existence of a "finance uncertainty multiplier": the impact of uncertainty shocks is amplified by worsening financial frictions.²⁹ Our evidence points to a possible "global finance uncertainty multiplier".

5 Related literature

Our paper contributes to the literature on uncertainty and the business cycle in several respects (for surveys, see Bloom (2014), Fernández-Villaverde and Guerrón-Quintana

²⁹The concept of a "finance-uncertainty multiplier" has been introduced by Alfaro, Bloom, and Lin (2021). Other theoretical models dealing with the interaction of uncertainty shocks and financial frictions are Gilchrist, Sim, and Zakrajšek (2014) and Arellano, Bai, and Kehoe (2019). Empirical evidence for the US has been documented by Caggiano, Castelnovo, Delrio, and Kima (2021). They run a counterfactual exercise that "shuts down" the response of a measure of credit stress to a financial uncertainty shock. An advantage of the exercise proposed in this paper is that it does not force any endogenous variable of our VAR to stay still after the GFU shock.

(2020), Cascaldi-Garcia et al. (2021), and Castelnovo (2022)). Our empirical findings on the recessionary effects of financial uncertainty shocks at a global level support the predictions of models featuring either external uncertainty shocks hitting open economies (Benigno, Benigno, and Nisticò (2012), Fernández-Villaverde, Guerrón-Quintana, Rubio-Ramírez, and Uribe (2011), Born and Pfeifer (2014), Mumtaz and Theodoridis (2015), Handley and Limão (2017), Chatterjee (2019)), or global volatility shocks (Gourio, Siemer, and Verdelhan (2013)). Our results - based on a world-level dataset - that global financial uncertainty shocks are relevant drivers of world output echoes those on the role of financial uncertainty for the US business cycle (see, e.g., Caggiano, Castelnovo, and Groshenny (2014), Leduc and Liu (2016), Basu and Bundick (2017), Ludvigson, Ma, and Ng (2021), Caggiano, Castelnovo, and Nodari (2022)), above all in periods of high economic and financial stress such as the great recession (Caggiano, Castelnovo, and Pellegrino (2017), Alessandri and Mumtaz (2019), Caggiano, Castelnovo, and Pellegrino (2021), and Pellegrino, Castelnovo, and Caggiano (2022)).

A relevant paper also dealing with the measurement of global financial uncertainty and the effects of global financial volatility shocks is Cesa-Bianchi, Pesaran, and Rebucci (2020). They propose a factor-augmented multi-country panel VAR model to estimate quarterly measures of global macroeconomic and financial uncertainty from a dataset comprising output growth and equity market realized volatility for 32 advanced and emerging economies. They find that the negative correlation between global financial volatility and world output growth is mainly driven by shocks to the latter variable. They also find that global financial volatility is mainly affected by its own shocks, and that such shocks have strong and persistent effects on world output. Cesa-Bianchi, Pesaran, and Rebucci (2020) also model country-specific output growth and financial volatility shocks, which we do not deal with in our VAR analysis. On the other hand, we employ a novel identification strategy to identify GFU shocks and separate them from a battery of other shocks that are relevant at a global level (US monetary policy shocks, oil supply shocks, global financial cycle shocks, and world output shocks). Also, Cesa-Bianchi, Pesaran, and Rebucci (2020) achieve point-identification (which enables them to provide punctual estimates on the business cycle effects of global uncertainty shocks), while we work with set-identification (which enables us to explore model uncertainty and draw a connection between global financial stress and global output loss). In spite of these differences, both papers find that global financial volatility shocks: i) induce a persistent and quantitatively sizeable response of global output; ii) were among the

drivers of the global great recession.³⁰ For other papers measuring global financial volatility, see Kang, Ratti, and Vespignani (2017) and Bonciani and Ricci (2020).

Researchers have also worked on global macroeconomic (as opposed to financial) uncertainty.³¹ Our focus on financial uncertainty is justified by: i) the recent empirical evidence for the US showing that uncertainty stemming from financial markets is likely to be an exogenous and relevant driver of the business cycle (Angelini and Fanelli (2019), Angelini, Bacchiocchi, Caggiano, and Fanelli (2019), Ludvigson, Ma, and Ng (2021), Fernández-Villaverde and Guerrón-Quintana (2020));³² ii) the availability of monthly data, which allow our GFU measure to feature uncertainty spikes that are precisely estimated and are likely to be informative for the identification of uncertainty shocks. Such spikes would be harder to identify with quarterly data.

Our narrative restriction approach to identify global financial uncertainty shocks closely follows Ludvigson, Ma, and Ng (2021), a paper concerned with US uncertainty shocks. Antolín-Díaz and Rubio-Ramírez (2018) also propose to focus on selected events to sharpen the identification of macroeconomic shocks in set-identified analysis. Our narrative restrictions are meant to be a primary source of information for the identification of uncertainty shocks, while Antolín-Díaz and Rubio-Ramírez (2018) work with such restrictions to further shrink a set of admissible models already identified with traditional sign restrictions. Moreover, we prefer a frequentist VAR analysis to a Bayesian one because our moment inequality restrictions lead to identified non-Gaussian shocks. As stressed by Ludvigson, Ma, and Ng (2021), this element greatly complicates the Bayesian approach, but it is straightforward to handle via classical estimation.³³

³⁰Other differences between our empirical approach vs Cesa-Bianchi et al.'s (2020) include the series we model (stock market volatility only for 32 countries in their case vs. three volatility series for 42 countries in ours); the investigation of the role played by first-moment global financial cycle shocks for the global financial volatility and real business cycle, which we pursue; and the fact that we are not required to assume the lack of economic dominance by any of the countries we deal with.

³¹Among others, see Berger, Grabert, and Kempa (2016,2017), Mumtaz and Theodoridis (2017), Redl (2017), Miescu (2018), Ozturk and Sheng (2018), Mumtaz and Musso (2021), Carriero, Corsello, and Marcellino (2019), Carriero, Clark, and Marcellino (2020), Dibiasi and Sarferaz (2020), Ferrara, Karadimitropoulou, and Tryantafyllou (2021).

³²The debate on financial vs. macroeconomic uncertainty as drivers of real activity in the US is an open one. For a contribution pointing to macroeconomic uncertainty as exogenous to the US business cycle, see Carriero, Clark, and Marcellino (2019).

³³Antolín-Díaz and Rubio-Ramírez (2018) also stress the difference between their Bayesian approach and the existing frequentist approaches (see pp. 2803-2084 of their paper). For contributions that identify uncertainty shocks at a country-level with the strategy proposed by Antolín-Díaz and Rubio-Ramírez (2018), see Redl (2020), Caggiano, Castelnovo, Delrio, and Kima (2021), and Rivolta and Trecroci (2021). For a recent contribution on Bayesian inference in VAR models with non-Gaussian shocks, see Petrova (2021).

6 Conclusions

This paper estimates a novel measure of global financial uncertainty (GFU) with about 38,000 financial volatility observations covering 42 countries and 5 continents. We take to the data a multi-level dynamic factor model via which we produce a global financial uncertainty (GFU) factor as well as regional and country-specific uncertainty measures. GFU peaks in correspondence with well-identified historical events, its highest realizations occurring during the Great Recession and the COVID-19 pandemic. Within the context of a VAR analysis also modeling global output and the global financial cycle, we set-identify GFU shocks by imposing a combination of narrative, sign, ratio, and correlation restrictions. Our VAR analysis estimates the output loss attributed to GFU shocks during and in the aftermath of the Great Recession to be about 13%. Also, our VAR evidence supports the possibility of a "global finance uncertainty multiplier", i.e., financial stress following an increase in uncertainty might have contributed to the deterioration of global output due to exogenous increases in global financial uncertainty. Our findings support swift policy interventions aimed at maintaining the credit market in check to limit as much as possible the recessionary effects of uncertainty shocks.

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<i>Event restrictions</i>			
t	Event	Restriction	% retained draws
1998M8	Russian, LTCM default	$e_{GFU,t} > P_{50}(\mathbf{e}_{GFU,t}(\mathbf{B}))$	11.21%
2001M9	9/11	$e_{GFU,t} > P_{50}(\mathbf{e}_{GFU,t}(\mathbf{B}))$	
2002M6	Worldcom, Enron	$e_{GFU,t} > P_{50}(\mathbf{e}_{GFU,t}(\mathbf{B}))$	
2008M9	Great recession	$e_{GFU,t} > P_{75}(\mathbf{e}_{GFU,t}(\mathbf{B}))$	
2010M4	Greek crisis	$e_{GFU,t} > P_{50}(\mathbf{e}_{GFU,t}(\mathbf{B}))$	
2011M8	Eurozone crisis	$e_{GFU,t} > P_{50}(\mathbf{e}_{GFU,t}(\mathbf{B}))$	
2007M10-2009M6	Great recession	$0 > \sum_{t=2007M12}^{2009M6} e_{WIP,t}$	69.07%
<i>Correlation restrictions</i>			
	$ \rho(e_{GFU,t}, e_{USMPt}) < 0.05$	$ \rho(e_{GFU,t}, e_{OILt}) < 0.05$	6.94%
<i>Sign restrictions</i>			
	$\frac{\partial WIP_t}{\partial e_{GFC,t}} > 0$	$\frac{\partial GFC_t}{\partial e_{WIP,t}} > 0$	19.49%
<i>Ratio restrictions</i>			
	$\left \frac{\partial GFU_t}{\partial e_{GFC,t}} \right < \left \frac{\partial GFC_t}{\partial e_{GFU,t}} \right $	$\left \frac{\partial GFU_t}{\partial e_{GFU,t}} \right > \left \frac{\partial GFC_t}{\partial e_{GFU,t}} \right $	32.86%
<i>Entire set of restrictions</i>			0.08%

Table 1: **Set of identifying restrictions.** Constraints imposed to identify global financial uncertainty and separate them from global financial cycle shocks, global output shocks, oil supply shocks, and US monetary policy shocks. GFU stands for global financial uncertainty; GFC for global financial cycle; WUI for World Industrial Production. The last column reports the percentage of draws meeting each subset of restrictions, as well as (last row) the entire set of constraints we impose to identify the GFU shocks.

<i>2-year horizon</i>			
Variable/shock	e_{GFU}	e_{GFC}	e_{WIP}
GFU	69.12% [53.23%,82.88%,]	9.02% [7.46%,11.04%,]	21.62% [9.41%,36.99%,]
GFC	30.73% [16.61%,63.45%,]	22.97% [15.60%,33.29%,]	44.85% [13.48%,66.30%,]
WIP	8.91% [1.92%,44.29%,]	10.11% [5.27%,18.56%,]	79.74% [47.75%,92.56%,]
<i>4-year horizon</i>			
GFU	63.35% [47.82%,79.44%,]	9.66% [8.05%,11.98%,]	26.94% [12.51%,42.43%,]
GFC	30.74% [16.59%,63.51%,]	22.90% [15.54%,33.20%,]	44.93% [13.51%,66.38%,]
WIP	9.34% [1.98%,45.15%,]	10.11% [5.24%,18.53%,]	79.37% [46.86%,92.51%,]

Table 2: **Forecast Error Variance Decomposition: 2-year and 4-year ahead horizons.** Forecast error variance decomposition conditional on our retained models. Figures refer to the median value across retained models of the contribution of each given shock for each given variable at selected horizons. Figures in brackets: Minimum and maximum contributions according to our selected models. The sum of the contributions of the shocks per each given variable does not return 100 percent because this Table considers percentiles of distributions across different models of our retained set.

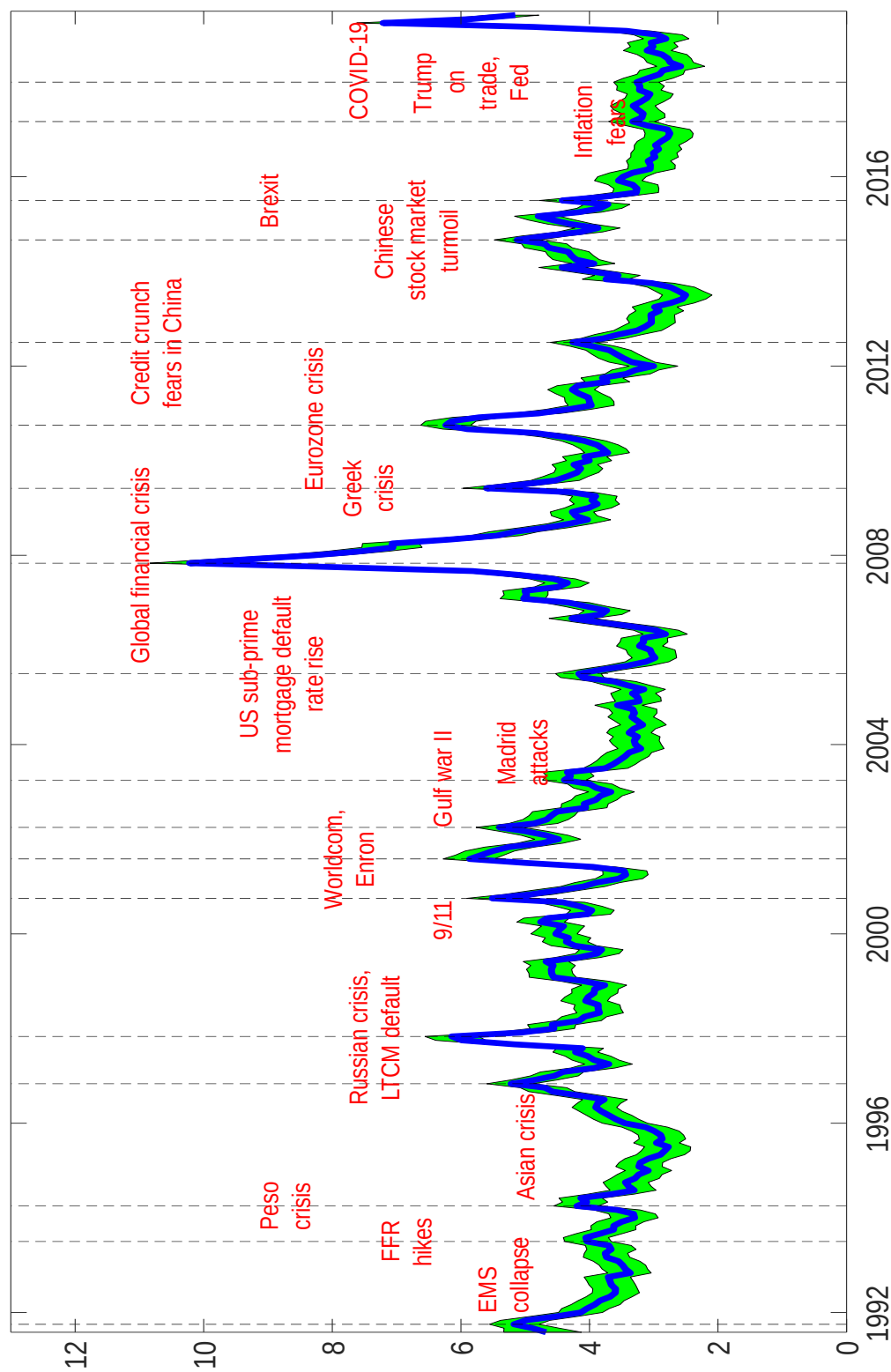


Figure 1: **Global Financial Uncertainty Factor.** GFU factor estimated with a dynamic hierarchical factor model à la Moench et al. (2013) combined with the algorithm designed by Banbura and Modugno (2014) to handle missing observations in an unbalanced panel. GFU is the global-level factor of the DHFM that jointly models regional, country-specific, and series-specific factors. Green area: 95 percent confidence bands regarding the estimation of the factors in the dynamic factor model.

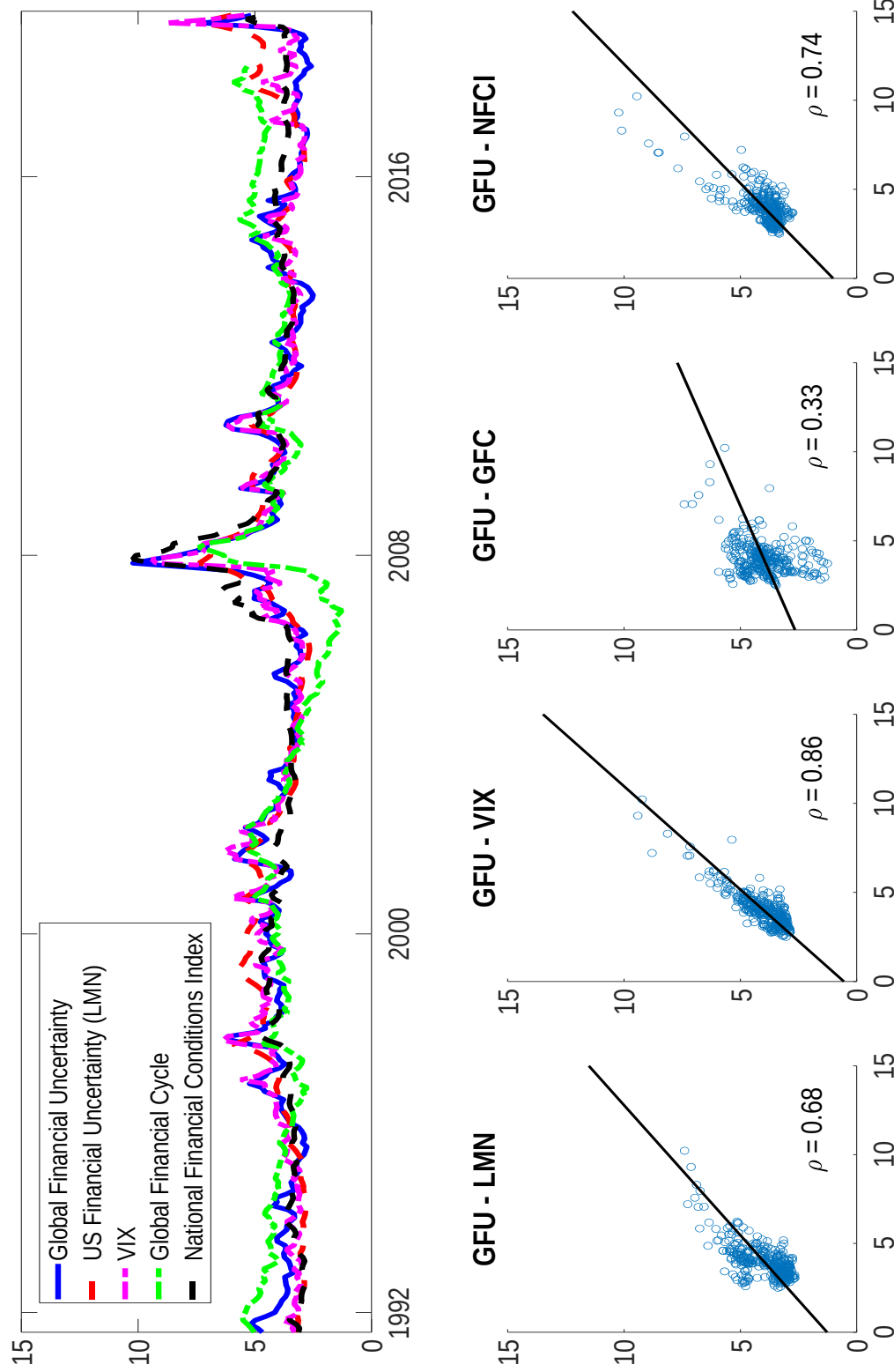


Figure 2: **Global Financial Uncertainty vs. Alternative Financial and Uncertainty Measures.** Upper panel: Global Financial Uncertainty (this paper) vs. Ludvigson et al.'s (2021) US financial uncertainty index, Miranda-Agrippino and Rey's (2020) global financial cycle index, the VIX, and the National Financial Conditions Index (NFCI). Lower panel: Pairwise correlations. All series in this Figure are normalized to have the same mean and standard deviation of the GFU series. Global Financial Cycle's sign (upper panel) flipped to ease comparability.

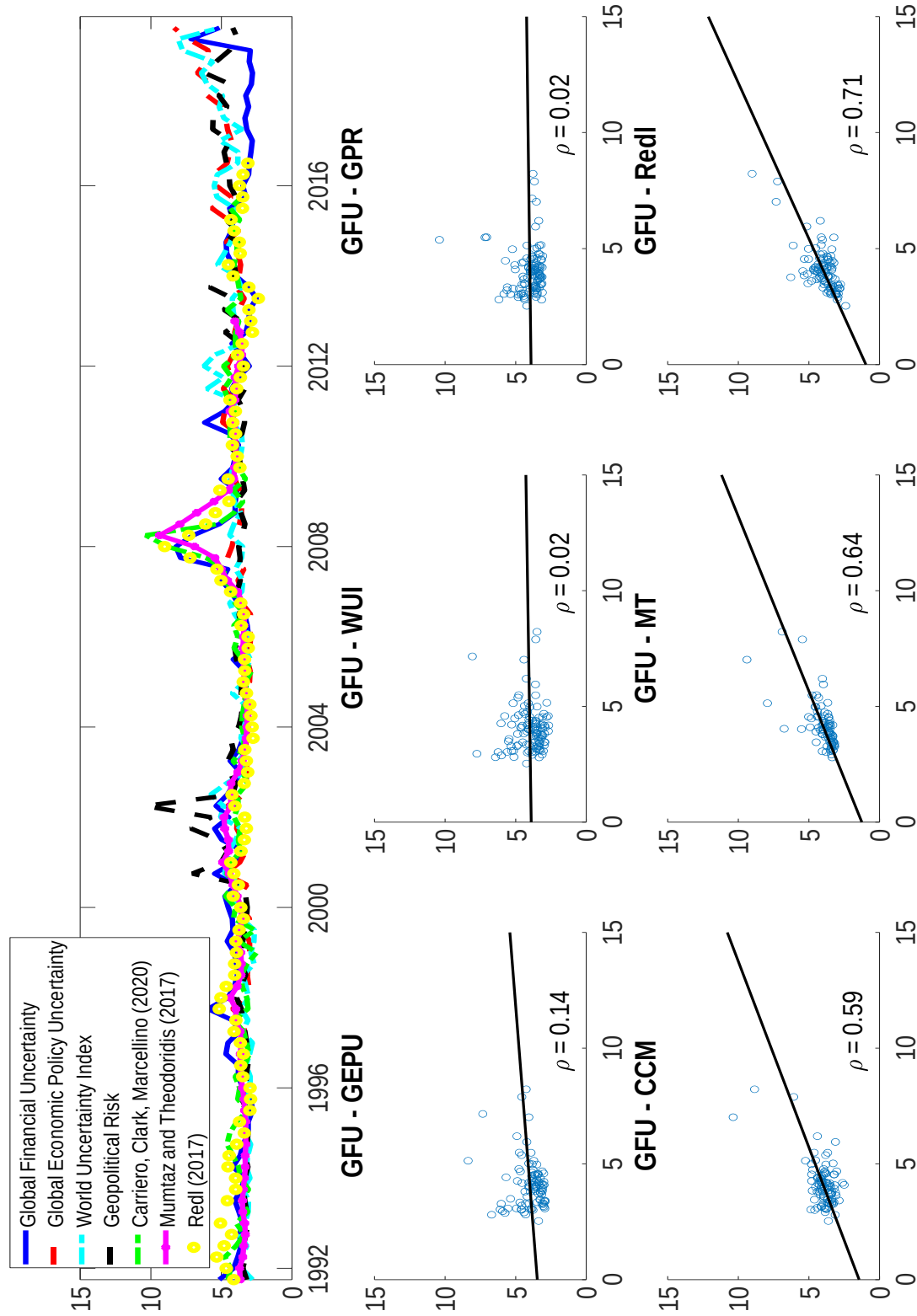


Figure 3: **Global Financial Uncertainty vs. Alternative Global Uncertainty Measures.** Upper panel: Global Financial Uncertainty (this paper) vs. Global Economic Policy Uncertainty as in Davis (2016), Geopolitical Risk as in Caldara and Iacoviello (2022), World Uncertainty Index as in Ahir, Bloom, and Furceri (2022), and Global Macroeconomic Uncertainty as in Carriero, Clark, and Marcellino (2020), Mumtaz and Theodoridis (2017) and RedI (2017). Quarterly frequencies. All series in this Figure are normalized to have the same mean and standard deviation of the GFU series.

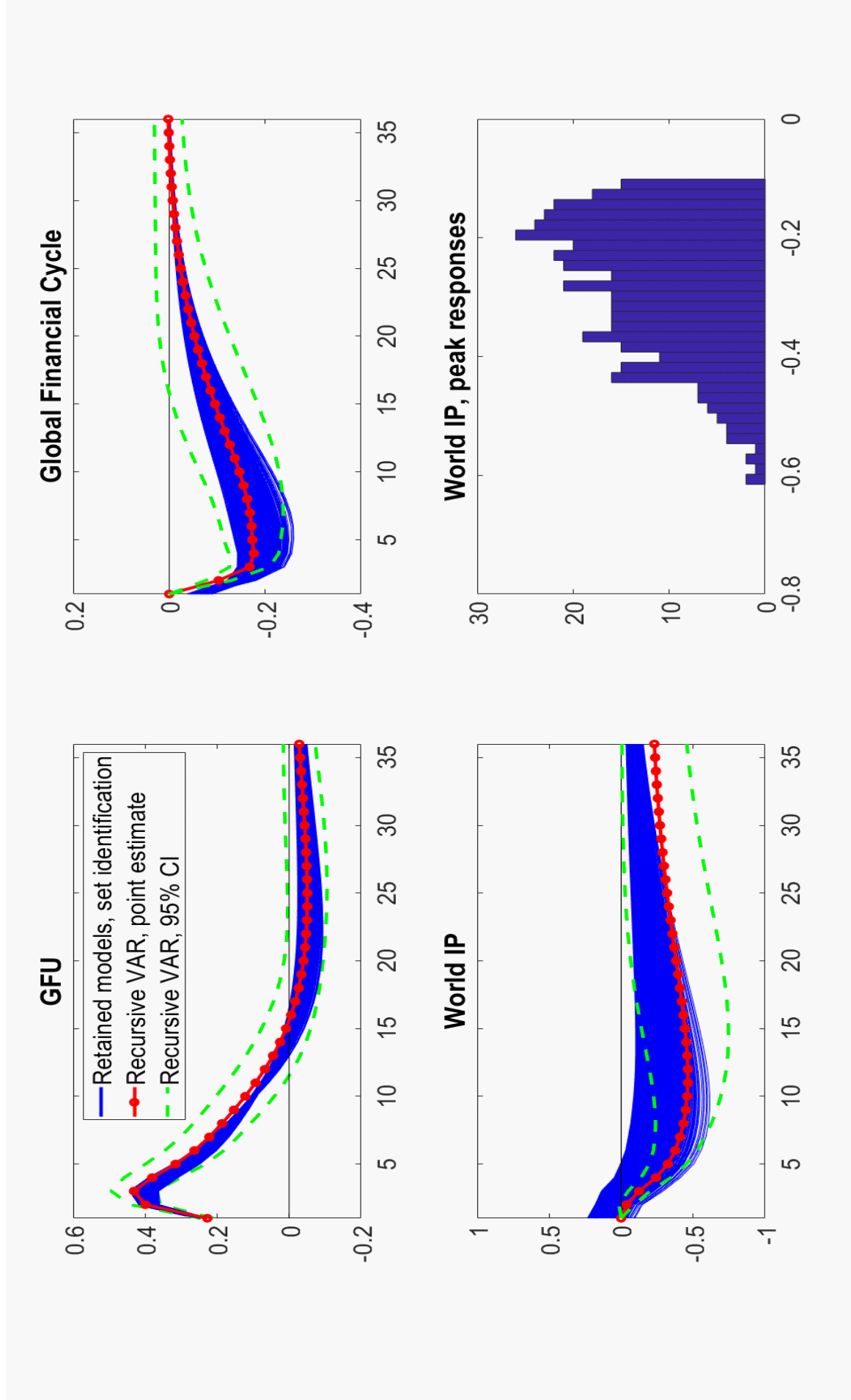


Figure 4: **Impulse Responses to a Global Financial Uncertainty Shock.** Sample: 1992M7-2019M4. Blue solid lines: Set-identified responses to a one standard deviation GFU shock. Identification achieved by imposing the restrictions reported in Table 2. Red solid lines with circles (green dashes lines): Point-identified responses (95 percent confidence intervals) produced with a recursive VAR with GFU ordered last. Bottom-right panel: Distribution of the peak responses across set-identified models.

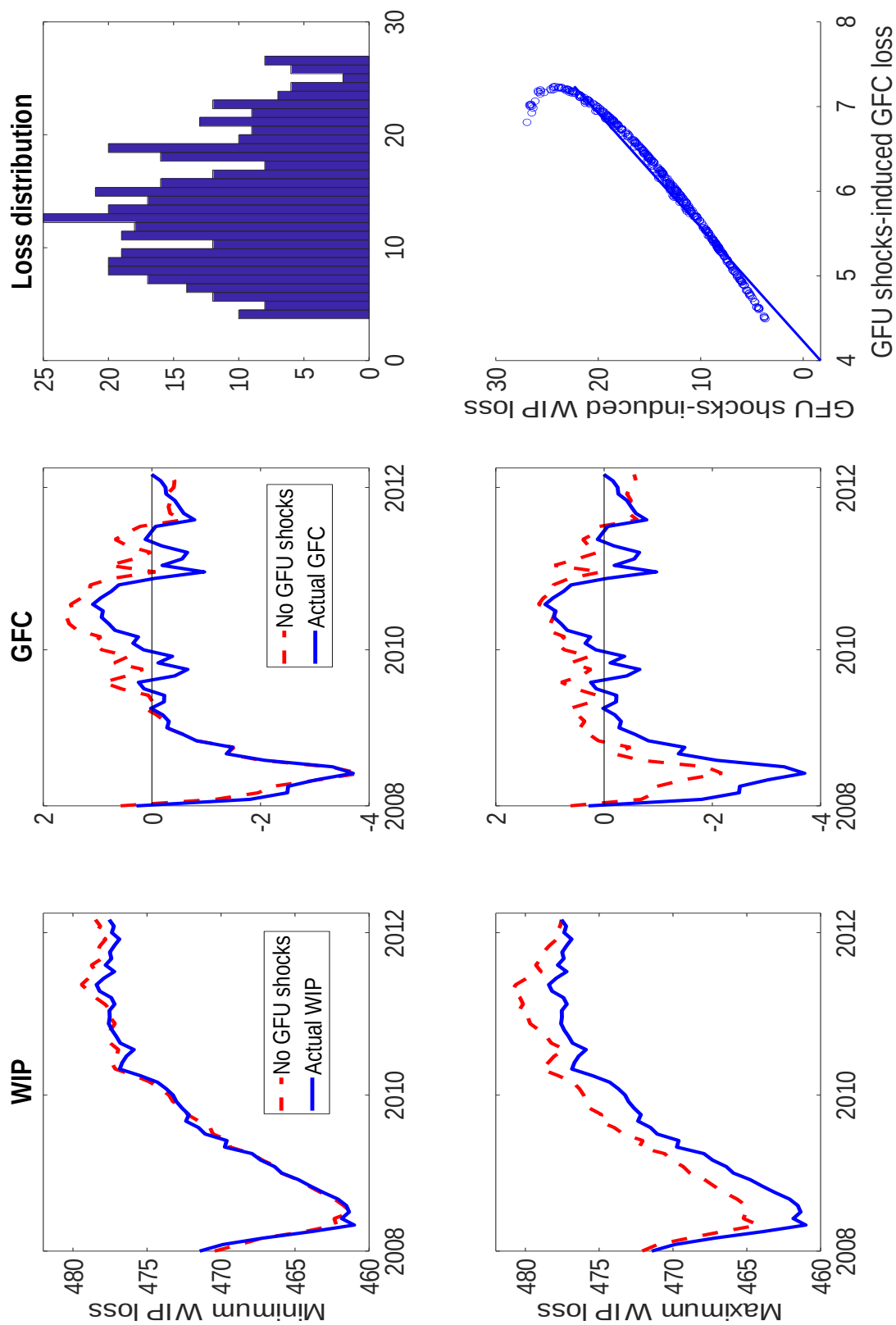


Figure 5: **GFU shocks and the Great Recession: Counterfactual simulations, WIP loss distribution, and the Global Finance Uncertainty Multiplier.** Sample: 2008M9-2012M12. Left and middle panels: Blue solid lines: Actual series. Red dashed-lines: Counterfactual series simulated with our estimated structural VAR by shutting down GFU shocks. Models associated with the minimum (upper left panel) and maximum (lower left panel) WIP loss plotted in the figure. Upper-right panel: Distribution of the WIP loss across all 406 retained models. Lower-right panel: Scatter plot correlating the World Industrial Production loss and the deterioration of the Global Financial Cycle due to GFU shocks according to all our retained models.